days volume in the closing auction compared to 4.0% on a normal day. There was no change in the volumes traded in the opening auction on index change days.

- *Month End*. Total volumes were 7% higher for large cap stocks and 9% for small cap stocks. Intraday volumes were only 3% higher for both large and small caps compared to normal days. Closing auction volumes were 8.6% of a normal day for large caps and 10.0% of a normal day for small caps. Opening auction volume was slightly less than what is normally experienced.
- *Quarter End*. Surprisingly, closing auction volumes, while higher than a normal day's, was actually lower than month end volumes. Large cap volumes were 5% higher and small cap volumes were 6% higher than normal days. There were no differences in intraday volumes compared to normal trading days. The big difference was similar to month end. Large caps traded 7.8% of a normal day's volume and small caps traded 8.4% of a normal day's volume in the closing auction.
- *Before/After Holidays*. Volumes were lower as expected. Large caps traded 8% lower and small caps traded 4% lower than on normal days. Intraday volumes changes were pretty similar. Intraday large cap was 9% and intraday small cap was 5% lower than on normal days.
- *Early Close*. Large cap volumes were −67% of normal days and small cap volumes were −52% of normal days. Closing auction volumes were also much lower, large cap was 0.9% of a normal day and small cap was 2.1% of a normal day. Opening auction volumes, however, were consistent with normal days.

Each of these categories has different repercussions for trading, traders, and algorithms and insight into each can result in dramatically improved trading performance. Conducting on-going market microstructure research for these special event days provides analysts and developers with essential insight into how best to transact stocks on these days. This could lead to improved trading performance and higher portfolio returns (Table 2.10).

FLASH CRASH

The flash crash of May 6, 2010 was a very significant financial event for many reasons. On the morning of May 6, 2010 the market opened slightly down from the previous close. The financial markets were worried about the debt crisis in Greece—potential defaults—which was dominating the news. At about 2:45 p.m. the flash crash hit. The SPX index lost 102 points